

Machine Learning and its Applications

Today's goals



Module 1

Machine Learning Fundamentals I: Basic Concepts

Module 2

Machine Learning Fundamentals II: Model Training and Evaluation

Module 8 Introduction

Introduction to Module 8's Study

Video • 1 min

What Does “Good” Mean in Machine Learning Models?

Reading • 5 min

Module 8-1 ML Model Training Fundamentals

ML Model Training Fundamentals: Parameters, Hyperparameters, and Loss Functions

Video • 8 min

ML Model Training Fundamentals: Loss Functions and Gradient Descent

Video • 5 min

ML Model Training Fundamentals: Gradient Descent

Video • 10 min

Materials for ML model training

Reading • 10 min

What Does It Actually Mean to “Train” a Machine Learning Model?

Reading • 5 min

Module 8-2 ML Model Evaluation Fundamentals

ML Model Evaluation Fundamentals: Fundamental Concepts in ML Model Evaluation

Video • 13 min

ML Model Evaluation Fundamentals: Fundamental Concepts in ML Model Evaluation

Video • 5 min

Materials on ML Model Evaluation Fundamentals

Reading • 10 min

When a Model Fails: Is It Bias, Variance, or Something Else?

Reading • 5 min

Module 8-3 Summary of the ML Process

Summary of the ML Process

Video • 5 min

Materials for ML Process

Reading • 10 min

From Data to Decisions: Reconstructing the Machine Learning Pipeline

Reading • 5 min

Module 8 Learning Coach Dialogue

• 5 min

Module 8 Quiz

Module 8 Quiz

Practice Assignment • 30 min

Module 8 Assignments

Module 8 Assignment 1

Graded App Item • 30 min

Module 8 Assignment 2

Graded App Item • 30 min

CSV file for Module 8 Assignment

Reading • 1 min

Module 8 Recap

Reading • 5 min

Module 3

Data Preparation

Module 4

Support Vector Machines

Module 5

Artificial Neural Networks

When a Model Fails: Is It Bias, Variance, or Something Else?



0:00 / 1:46



2x





1. Reflecting on Training vs. Test Behavior

- Reflect on what it means when a machine learning model performs poorly on both training and test data. What might this suggest about model complexity, bias, or the model's ability to learn meaningful patterns?
- Reflect on situations where a model performs very well on training data but poorly on test data. What does this behavior suggest about variance, overfitting, and generalization?

2. Reflecting on Sources of Prediction Error

- Reflect on why prediction error is often considered a combination of bias, variance, and noise rather than a single quantity.
- Consider which parts of prediction error can realistically be reduced through better modeling, data preparation, or training strategies, and which parts may be unavoidable due to randomness or uncertainty in the data itself.

3. Reflecting on Improving Generalization

- If a model is underfitting, reflect on the types of actions that could improve its learning capability. For example, consider model complexity, feature engineering, training duration, or data representation.
- If a model is overfitting, reflect on the actions you could take to improve generalization performance. Why might techniques such as regularization, cross-validation, feature reduction, or collecting more data help?

4. Reflecting on Evaluation Strategies

- Reflect on why relying on a single train/test split may sometimes give a misleading picture of model performance.
- Consider how k-fold cross-validation may provide a more reliable estimate of model quality and robustness across different subsets of data.
- Reflect on situations where comparing models without cross-validation could lead to unreliable or unfair conclusions.

Mark as completed

-  Like
-  Dislike
-  Report an issue